Unit 2 - Computer Systems
Assignment 5 - Upgrade Software

Introduction

You are newly appointed computer technician working for a Jersey Bank. You have been asked to prepare for an installation, upgrade and planned maintenance tasks. You are required to complete the following tasks to demonstrate your skills and knowledge in this area.

Task 1

(P1) Describe the potential prompts that initiate the installation of new or upgraded software

Task description:*An organisation will also need to change their IT systems, installing new or upgrading existing software. Under each of the following headings give a detailed example of situation that would initiate the need for new or upgraded software:*

Problems with existing systems

Throughout their time and use, PCs and other systems can go outdated and slower than a newer of brand new PC, which will have an effect on the performance of the PC, making it necessary to upgrade the new software and hardware within the system. The software that is already installed in the PC is likely to be out of date or in need of an upgrade in order for it to do its full potential. It is certain that an upgrade to a PC would improve the performance of the computer compared to what it previously was, making the workload much more efficient.

There are other problems that could also develop with lack of upgrading, which would be that the software is unstable which would cause the PC system to crash and harm it. However, this is also a risk with installing a new software. The reason for this is that the software might be old and out of date and newer versions of it might be available for installation so that it could be much more supported and efficient.

It's also important to keep in mind that if any change was made to the software, there is a chance it can have a negative impact on the system. For example, some of the applications will be corrupt and will stop working, bugs can develop within the software itself which also causes corruption or that the system will not accept the changes.

The upgrade of Operating Systems are also very important due to their importance on a PC, it makes a very significant difference of which OS is installed on the PC. Windows 8 / 8.1, Windows 7 and even Windows XP are very popular OS within a company. Also, with the release of the New Windows 10 OS, an upgrade should be considered, especially with an old OS.

The Windows 8 /8.1 OS did improve from the older versions of the Windows OS, especially with newer software compatibility and security. However, Windows 8 did suffer from performance issues for some PCs due to their hardware being too old to handle the newer OS, which means that a PC will need to be upgraded to be compatible with Windows 8's performance. However,

with a better PC, Windows 8 /8.1 would run better than an older Windows OS. User efficiency is also a big problem for Windows 8, due to the vast UI change. This would leave the employees of the company to be frustrated due to the obnoxious change and need a training programme to teach them how to use the new OS, which would cost the company more.

The Windows 7 OS was a very good OS, but it is now getting outdated and its loosing its software compatibility with newer and more efficient software, technically reducing the efficiency of the OS. Windows 7 also lacks a huge performance when compared to newer OS such as Windows 8.1 and Windows 10, which means it runs applications slower.

Windows XP was also a very good OS, and there are businesses and organisations that still use Windows XP. However, it has a massive compatibility problem due to its discontinuation by Microsoft in 2014. Which means that PCs that still continue using Windows XP will be very vulnerable to security risks and viruses. Especially when browsing through the discontinued Internet Explorer 8, which would expose the PC to many threats. Windows XP also has a lot of apps which will not work with the OS, due to the software and hardware manufacturers also discontinuing the support for Windows XP.

The Windows 10 OS is the newest OS made by Microsoft Windows, and it has a “improved” from the previous OS by Microsoft. However, this does not mean that Windows 10 doesn’t have any problems. The biggest problem is if you have an outdated PC in the company and upgrade it to Windows 10, which would cause the OS to run very slow because of the outdated hardware on the PC. Which means that the PC will need to be upgraded if Windows 10 was to be running properly and without frustration, costing the company a lot more money.

Additional functionality required

If an upgrade to the software is planned, then it is possible that the upgraded software would include extra features to use. This is one of the main reasons of upgrading a software, the other reasons being the improved performance to the software and that more work can be produced efficiently with the new features.

Newer software have become a lot more supportive of the functionality and support of the newer, modern technology that has proven to be much more useful for some users. For example, touchscreen was a very controversial invention that improved the ability of functionality and efficiency on older hardware, such as phones to smartphones.

Touchscreen functionality on newer OS like Windows 8 / 8.1 and Windows 10 may not be useful for most people, but it can be very beneficial to some companies. If the business relies on producing graphics design then the newer OS should be considered in an upgrade, but if a business doesn’t really need this sort of functionality, then an upgrade shouldn’t be needed as much or at all. With software application compatibility, the newer OS should be the best OS to upgrade to because of the prioritised support the software manufacturers give.

This means that the software that a business or organisation uses to produce a more efficient workload for the company, proving the upgrade possibly profitable.Customisation of the Operating System may not seem very important for a business point of view, but for an employee it can be beneficial due to the fact that the better customisation of newer OS will

make it a much more comfortable workspace for them, which has a possibility of the employee doing the work better than an OS with a fixed look and little to no customisation which would appeal to them.

New hardware requiring new or upgraded software

During the installation planning of hardware, it is important that a device driver is installed within the computer system so that the hardware that is about to be installed can be controlled accordingly. If the newer hardware was installed within a PC, it is likely that it can have a negative effect on the software or the other bits of hardware. THis can be due to the newer hardware not being compatible with the software or the other pre installed hardware in the PC, and an upgrade is the only chance of making it work properly.

With newer software and Operating System supporting 64 Bit, which can make a massive difference if you have a PC with decent hardware. This may be extremely important or not very important for businesses, it depends on what type of business it is that is considering an upgrade. If it's a business which works with graphics design or developing games, then getting a 64 Bit system should be a very big priority for them. If it's a business that sells goods towards the public or it's a financial business, then a 64 Bit system would be a waste of money because the upgrade would provide little to no difference.

If a business was to upgrade their OS, they will also need to ensure that the compatibility of the other updated technologies are compatible with the OS that they have upgraded to. It's likely that the newer software and technologies are compatible with the newer OS, but there is still a risk that the software and technologies may not be compatible with the newer OS and if that problem wasn't noticed initially, it would cause big problems for the business or organisation that was affected.

External prompts for software bug fixes

It is important to check the new software before even considering an update, because of the amount of possible risks. For example, there is a possibility that the software would have some soft of software bug or other corruptions due to the manufacturer not addressing or noticing the software bugs. This means that the software may work initially, but then bugs will soon be noticeable and would cause the software to crash and it will prevent from working until a new update that addresses the issue is released.

Before installing new software or hardware, it is recommended to follow the orders that the software or hardware manufacturer would prompt so that updates can be carried out whenever there are new updated with the software or hardware, which can improve its security, preventing from the important files from being lost or stolen. If there is a significant fault within the software or hardware it would need to be addressed as soon as possible due to its significant damage potential towards the computer system, such as crashing or even data loss.

Software and hardware bugs and problems are not only occurred within the system itself, sometimes there can be changes within the law and the software may not be supported, which leads to the software being discontinued. This means that an upgrade would be necessary if this were the case. With a discontinued software, it would either mean that it will have no more

updates or it will be no longer compatible. Without any updates to the software means that it would have a higher security risk, which means that the work has a higher chance of being lost or stolen.

Company policy (maintaining competitive advantage).

It is a requirement for a business or organisation to have policies so that the software used on the company can be under control. The way that is controlled is on one machine, which is the administrator.

This is so all the machines within the company can all have the same software to work with and that it stops the employees from installing other unneeded software, because it might be illegal or it might have a virus which can be harmful to the system or even the company. If additional upgrades or installations are needed, a plan would need to be made to everything in order for the process to be in a controlled manner.

It is very likely that a company would want the latest version of the software and Operating Systems due to the latest and more efficient features that they come with, so that the employees within the company can produce a lot more work than previously and possible with a much faster pace.

However, a company may also be forced to upgrade or purchase a software or Operating System, due to it being extremely necessary for the software that is used within the company or that the company needs new functionality from the upgrade in order to fully fulfil the purpose of the business, and of course the software has to be legal and relevant towards the company's needs.

Task 2

(P2) Describe the potential risks of installing or upgrading software

Task description:*When upgrading existing software problems can occur and affect the organisation. Under the following headings give a detailed example of the risks involved and the affect it could have on the organisation.*

Potential loss of service

When the computer system is in the process of installing new software, it's on a time gap which is commonly called “down-time”. This is when a computer is inactive due to the software that is being installed using the essential parts of the PC in order for it to install properly.

The installation of the software can last to about an hour, but this is a very inaccurate measurement because all softwares have different files sizes, it also depends on how much performance the PC systems have. This can be very time consuming, especially for a business point of view, because of the missed out workload and profits for the business. For example, if the business was interrupted due to the upgrading of an operating system, it would restrict the internet access and other network resources until the installation is fully completed.

Incompatibility issues

As the software is being installed, it is important to always make sure if the PC and the software is compatible with each other. If the PC or the software weren't compatible with one another, it is highly likely it will not install. If it does install, then there would be multiple significant technical issues with the software running on the computer system, i.e. the screen will turn off and stal black, inconsistent software flow (crashing), possible delete or corrupt files on other programs within the system or it could put the computer at risk of repairs.

With installing an Operating System, it is extremely important to backup the whole hard drive so all the software are work contained doesn't get lost. For example, if Linux was to be installed in a Windows computer and the Windows computer wasn't backed up, then all the files within the Windows OS will be deleted and replaced with the new Linux OS and it would be impossible to retrieve the old files back.

Staffing Issues

When installing new updates or software there undoubtedly are risks in carrying them out, and even if the risks were properly addressed and everything was installed without flaw. There is still a problem that can occur with the new and upgraded machines, which is the staff.

The big risk of the staff is that they will be very unfamiliar with the new upgrade. To reduce this problem is to add further training, so that they can use the software's new features. This could be very impactful towards the business due to the loss of valuable working time and the cost of training to every employee that has had their computer system upgraded.

Sometimes, the software may not be likes in general. This is due to the personal taste in the employees. There are many reasons for this, it could be because the software could be difficult to use or goes about doing certain things completely different to than it did previously. This may result in people refusing to use it or making life difficult for those who run the software or can also demotivate employees in the workplace causing them to do work at a lower standard which is bad for the company.

One method of reducing this risk in the first place is too look for a new software which is somewhat similar to the previous software, but it's very unlikely. An example of this would be an upgrade of an OS. If the computer systems have Windows 7 installed in them, it would be best to install Windows 10 instead of Windows 8 due to the fact Windows 10 is a lot more similar to Windows 7, more similar than Windows 8 is to Windows 7.

Risk reduction measures

When carrying out an upgrade or installation, it is essential that there is an understanding of the risks involved with any installation or upgrade and to have measures in place to reduce the risks as much as possible.

The first measure of reducing the risks is backups, which basically a copy of the files of data within a harddrive so it can be used in case the original data gets damaged or lost. It is essential that before making any installation attempts that the data and software should be backed up

within the computer systems.

The reason for this is due to the installation having risks of damaging existing data and software. With backed up data, it could easily replace the damaged or corrupt data. Without backed up data, all the damaged data and even the data connected to the damaged one, would be lost and unable to work.

The second measure of reducing the risks is installing at low risk times. This is very important, the reason being that if a business installs a software or update during the busiest hours, it could be very time consuming due to the large amount of time it takes to install the new software, consuming very valuable business time.

Instead of doing it during the busiest time, it would be best to do it during the weekend if the business has their employees off during the weekend. If that isn't the case, then doing the upgrade on night time is the best method, so that it is safer if there was a problem because it would take longer to repair and maintain the problem during a less busy time, instead of costing the business valuable working time.

The third and final measure of reducing the risks is doing an incremental installation, which is basically “testing” the installation of the software or a test PC. For example if a business is going to install a new software on multiple computer systems, it’s highly important not to install on all the computers at once. The reason for this is due to the risks that might occur during the installation, and if it is installed on only one computer system instead of multiple, it would be significantly faster to repair or discard the upgrade without damaging or corrupting files on all computer systems within a business.

Task 3

(D2) Evaluate the risks involved in the installation or upgrade of software and explain how the risks could be minimised

Task description: *Your managers are concerned about the risks involved with the upgrade of the Windows 8.1 to Windows 10*
You will need to:

1. *Identify and describe the risks involved to the organisation of the upgrade, stating how important to the smooth running of the organisation they are.*
2. *Identify and describe what measures can be taken to reduce or eliminate these risks.*
3. *Conclude with a summary of whether the risks can be controlled and whether it would be worth proceeding with the tasks*

The risks should involve such areas as potential loss of service and incompatibility issues.
The risk reduction measures should include backups and choosing appropriate low-risk time for installation.

Description of risk associated with the upgrade

Loss of Data

Description of Risk

Undoubtedly, data is the most precious thing about a computer system. With the Operating System and the applications containing and manipulating that data, this makes the OS and software very important for a business. However, all this can be upgraded and reinstalled, which can make the data vulnerable to risks such as damage and corruption which would make the data inaccessible. SInce data is often unique, it is impossible to ever retrieve the old data back.

Description of Control

The risks that could occur with the loss or damage to data can be minimized so it has little to no impact to the company. There are a few ways of minimizing these risks, which include:

Creating a backup file, which is a copy of all the stored data within the computer system so it could replace the original one if it were to be damaged or lost. However, it has to be created before any changes are being made (such as an Operating System upgrade), because if something goes wrong and you back it up, that means that you will have two copies of damaged or corrupted data and the data that works would be lost.

Low-risk time installation which is basically choosing the time carefully for the installation. This means that everything has to be planned before, so no time is wasted. It can't be done on day-to-day running of the business, it should be considered to do it during the weekend instead of the working days, or maybe even during the holidays.

Pilot deployment which is basically using one test PC as a test for the software if it has anything wrong with it. So if something does go wrong, only the test PC will be affected.

If all the above doesn't work, a System Recovery Point could be carried out. Which is basically a screenshot of the computer's Operating System is taken prior to the installation. So if the installation does go wrong, this is going to be used to reverse the damage caused.

Evaluation of Risk against Control

If anything were to go wrong with the upgrade of the Operating System and the computer system stops working, the backup file can be reinstated and it would have the older, original version working fine. This proves to save a lot of valuable time and data due to the fact that a backup file is easy to make and it's fast to replace the current data if it were damaged. You can also choose where you want the backup file and how you want to save it.

I believe that the advantages of the control outweigh the problems associated with the risk because a backup file is basically a copy of the original so it would work 99% of the time. However, there is still a very miniscule risk that the original data may not work even after

replacing the damaged data, but since this risk is so low there is no need to worry much about it. If you were to still to worry about that small risk, another backup file can be made just incase the first one doesn't work, which halves the risk.

If the backup does go wrong during a low-risk time installation, it means that repairs to the damage could be done without affecting the valuable work time, which means that the business wouldn't be losing work time or money due to the maintenance.

To reduce of all the risks of above not working, the Pilot Deployment would mean that only one PC would take all the damage if the risks weren’t contained properly, so it wouldn't have as much of an impact to the business as if it did with all the computer systems.

WIth the system recovery point, it is basically the concrete method on reducing the risks of losing all the data, which is basically a photograph of the contents of the Operating System, so that all the previous files could be re-installed.

Loss of Service

Description of Risk

If a computer system isn’t available for use after the new software has been installed, it is extremely likely that the company would stop its work, at the very least the work produced on a computer system. This means that the company would have to spare the precious work time in order to resolve the issues that were produced on the upgrade.

For example, if a bank’s website gets shut down, it wouldn’t be accessible by the customers so they can't find any information on their bank accounts, their bills or just about the bank itself. This could risk the potential loss of customers and the company would have to apologize to the customers for the technical difficulties that occurred.

In order to fix this you would have to download a patch or install new software, which could consume valuable business time. If you don't download a patch or install a new software, it could end up with the system going down either for a short or a long period of important time until the problem is solved.

There is a severe chance of this happening, because most the time the new software contains bugs or needs patches in order to work properly. Depending on the scenario, if this does happen to a business it would have a huge impact on it, which reduces the amount of work produced by the business, causing the business to lose a lot of time and money until the issue is resolved.

Description of Control

However, there are also ways to reduce this risk, for example:

A backup copy prior to the installation of all the data and software meaning that it can all be reinstalled if anything goes wrong leaving that no valuable information is lost and a lot of time will also be saved. When doing a backup, appropriate media should be chosen and it's essential that there is an understanding of what is about to be carried out, therefore it should be a planned event.

Another way is to use Pilot deployment, which allows for the old system to still be running in other departments but also being able to try the software at the same time. Pilot deployment allows for the old system to still be running in other departments but also being able to try out new systems at the same time, and it also allows for a comparison between the old and new systems.

Evaluation of Risk against Control

With the methods of reducing the risks, it is possible that it can reduce the risk of the organisations data as only a limited part of the organisation will be affected if the new software goes wrong.

Whether the advantages outweigh the risks is certain enough to still consider the upgrade from Windows 8.1 to Windows 10.

The Backup file is very advantageous in this situation due to the backup basically being a copy of the original so it would work 99% of the time. However, there is still a very miniscule risk that the original data may not work even after replacing the damaged data, but since this risk is so low there is no need to worry much about it.

The pilot deployment also allows for the panic in the company to be reduced with less people being affected by the change. Since only one PC would take all the damage, the risks that weren't contained properly wouldn't have as much of an impact to the business as if it did with all the computer systems.

Incompatibility

Description of Risk

There is a risk of the new Operating System not supporting the software that is currently installed to the computer systems, this means that the new operating system will have a chance of denying the access to some of the pre installed files. This can happen due to the software being out of date, or just a software bug.

The likelihood of this happening is quite low today, if you are keeping up with the newest software they will prompt you for the updates needed to keep your system compatible with the software but if a PC is running an old operating system and start downloading new software, it is possible that a few problems may appear due to the incompatibility.

Description of Control

There are a few ways in which the risks can be reduced. For example, one method is producing a backup copy prior to the installation of all data and software meaning that it can all be reinstalled if anything goes wrong leaving that no valuable information is lost and a lot of time will also be saved. When carrying out a backup, the appropriate media should be chosen and it is essential that there is an understanding of what is going to be done.

Another way is to get a few workers to test the software before installing it to every computer initially. Pilot deployment allows for the old system to still be running in other departments but also being able to try out the new system at the same time. This also allows for a comparison between the old and the new system.

Lastly, another way to reduce the risk is by doing a system recovery point by taking a screenshot of the computer's operating system prior to the installation, incase the installation doesn't go as planned and this will allow the ability to reverse the damage a lot faster.

Evaluation of Risk against Control

Incompatibility is a very massive risk because whenever installation of new Operating Systems doesn't match the hardware and conflicts with it, boot prevention is a risk that is very possible on happening and the only way to fix that is to restore point in startup repair screen.

The advantages of the use of pilot deployment is a great alternative to prevent mass hysteria in case the software is unsuitable for the computers only a specific department will be affected and prevented from working.

Together with back up points and pilot deployment, use of low risk time installation may smooth everything out in the case of possible problems by completing any required repairs on time and not using up the valuable work time.

Staffing

Description of Risk

It is possible that the staff may not like the new software that was installed, especially something as big as an Operating System from Windows 8.1 to Windows 10, which has a major User Interface change. This can result in people refusing to use it or making it difficult for those who

run the software, it can also demotivate employees in the workplace causing them to do work at a much lower standard which isn't good for the company.

The likelihood of this happening in today's society is quite high as most people do not like change and by trying to change that way things are done is quite minimal, with the only thing being affected is the business as when people do not like the software, this can ruin the reputation.

Description of Control

However, there are ways in which this can be reduced, for example: One way is by using a group of workers to test the software before installing it to all computers instantly. This pilot deployment can also allow the panic that can be produced within the company to be reduced with less people being affected by the vast change.

This way you can see if the group that has been chosen like the software and get their opinions on it, allowing a much more safer choice on whether to install the software on all systems or not to install it at all.

Evaluation of Risk against Control

It is very uncertain that the control would outweigh the risks when we are considering a new software installation due to most people not liking change, which means the difference of completely installing the new Operating System or completely abandoning the idea of installing the new OS.

If it were the case that the employees do not like the change with the new OS, the backup file could still prove useful due to the fact that the backup restored the OS to its previous state in which the employee feels comfortable with.

However, if there is a mix of employees that do like the new operating system and others who don't. There is the ability of backing up one specific PC, which means that it is possible to have one PC with Windows 8.1 and other with Windows 10, which would appeal to the employee.

Summary of the risks, how to control them, how effective those controls would be and whether your organisation should go ahead with the upgrade

In summary, whenever the software you use has an impact on the job or even the whole company. Therefore it is important to understand that upgrading or installing new software can be very risky and could turn out as a disaster. However, there are a few main factors that often get affected by unsuccessful software upgrade which are Loss of Data, Loss of Service, Incompatibility issues and Staffing issues.

There are many ways in which these issues can be compensated, making the software upgrade a lot less risky compared to what it was without the risks. The most common ways of reducing the risks are Backups, low risk time installations, pilot deployment and a system recovery point.

With backups, it is the safest way to protect the data from permanent deletion, corruption and unauthorized modification. To make sure that you get the best results it to prepare schedules back-ups. This is so that the PC backs up the stored data automatically, it is also best to have an additional internal drive or an external drive to save the backups.

Whenever software is being installed there is no way to exactly know if everything will go well, so it is always safe to back everything up before making changes so you could restore the lost data in the installation progress.

With low risk time installations, a company must always make smart decisions when it comes to assigning time frames for new software installation or upgrade. Otherwise, if anything does go wrong, the installation time will overlap with the work time of employees who needs a computer in order to work and will cost the company to lose valuable time and money due to the lack of service provided.

It is best to assign extra time for software installation process in case anything goes wrong, so that technicians would be able to sort everything out in time.

With pilot deployment, making software upgrades or getting a new software only for a single department is a smart decision as if anything goes wrong only one department will be affected and not the whole company and therefore mass panic will be reduced.

Installing software only within one department also gives the technician's knowledge of what software they should stick to for the whole company as they have the opportunity to make performance tests on the computer systems.

With a system recovery point, it is very similar to a back up. Basically, a screenshot of the previous state of the computer without any changes added to the computer system. So if anything were to go wrong with the installation or the upgrading process, you can undo it by simply restoring the system to the latest point before installation occurred.

However, this isn't as reliable as backups are because recovery points may sometimes not include additional software because the latest recovery point may be older than intended.

In conclusion, the upgrade is worth it due to the compensation of the risks and that the new Windows 10 OS will be compatible with much more modern and efficient software that would increase the amount of work produced, making the upgrade very profitable towards the company and I believe that the money wasted on the Windows 10 upgrade would be compensated to at least a break even.

However, the employees still risk the ability to keep and maintain this software. Even though there is a chance that almost everyone will accept the Windows 10 change but it is very uncertain, because the employees each have their own views on a new OS. Some would love a new Operating System and would beg for it, but there are some who are comfortable where they are and would hate the change from their comfort zone.

In conclusion, upgrading the software is worth it on a business point of view because of the more work that could be made due to the features of Windows 10 and its compatible software, making the upgrade very profitable. However, the employees will either love it or hate it, most likely have a mixed view on it. Basically, Its an opinion between company profits or employees comfort zone.